Final Exam Topics Overview

Final Topics

While only the content in the uploaded lecture notes/slides is expected for the final, textbook references and links to other background material are provided for each of the topics.

(*) indicates that the topic will not be tested on the final exam.

General Linguistics Concepts (Parts of J&M 2nd ed. Ch 1 but mostly split over different chapters in J&M. Ch.1 not yet available in 3rd ed.)

- Levels of linguistic representation: phonetics/phonology, morphology, syntax, semantics, pragmatics
- Ambiguity, know some examples in syntax and semantics, including PP attachment, noun-noun compounds.
- Garden-path sentences.
- Type/Token distinction.
- Know the following terms: word form, stem, lemma, lexeme.
- Parts of speech:
  - know the 9 traditional POS and some of the Penn Treebank tags (but no need to remember the complete tagset).
- Types of Linguistic Theories (Prescriptive, Descriptive, Explanatory)
- Syntax:
  - Constituency and Recursion. Constituency tests (*).
  - Dependency.
  - Grammatical Relations.
  - Subcategorization / Valency (and relationship to semantic roles).
  - Long-distance dependencies.
  - Syntactic heads (connection between dependency and constituency structure).
  - Center embeddings.
  - Dependency syntax:
    - Head, dependent
    - Dependency relations and labels
    - Projectivity

Text Processing (Split over different chapters in J&M. Parts of J&M 3rd ed. Ch. 6

- Tokenization (word segmentation).
- Sentence splitting.
- Lemmatization.

Probability Background

- Prior vs. conditional probability.
- Sample space, basic outcomes
- Probability distribution
- Events
- Random variables
- Bayes' rule
- conditional independence
- discriminative vs. generative models
- Noisy channel model.
- Calculating with probabilities in log space.

Text Classification (J&M 3rd ed. Ch. 6

- Task definition and applications
- Document representation: Set/Bag-of-words, vector space model
- Naive Bayes' and independence assumptions in text classification.

Language Models (J&M 2nd ed. Ch 4.1-4.8, J&M 3rd ed. Ch 4)

- Task definition and applications
- Probability of the next word and probability of a sentence
- Markov independence assumption.
- n-gram language models.
- Role of the START and END markers.
- Estimating ngram probabilities from a corpus:
  - Maximum Likelihood Estimates
  - Dealing with unseen Tokens
  - Smoothing and Back-off:
    - Additive Smoothing
    - Discounting
    - Linear Interpolation
    - Katz' Backoff
- Perplexity

Sequence Labeling (POS tagging) (J&M 2nd ed Ch 5.1-5.5, J&M 3rd ed. Ch 10.1-10.4

- Linguistic tests for part of speech (*).
- Hidden Markov Model:
  - Observations (sequence of tokens)
  - Hidden states (sequence of part of speech tags)
  - Transition probabilities, emission probabilities
  - Markov chain
  - Three tasks on HMMS: Decoding, Evaluation, Training
    - Decoding: Find the most likely sequence of tags:
      - Viterbi algorithm (dynamic programming, know the algorithm and data structures involved)
    - Evaluation: Find the probability of a sequence of words
      - Spurious ambiguity: multiple hidden sequences lead to the same observation.
      - Forward algorithm (difference to Viterbi).
    - Training: We only discussed maximum likelihood estimates.
  - Extending HMMs to trigrams. (*)
- Applying HMMs to other sequence labeling tasks, for example Named Entity Recognition
  - B.I.O. tags for NER.

Parsing with Context Free Grammars (J&M 2nd ed Ch. 12 and 13.1-13.4 and 14.1-14.4 and Ch. 16,
J&M 3rd ed. Ch. 11.1-11.5 and Ch. 12.1-12.2 [Earley not covered in 3rd. ed]
and Ch 13.1-13.4

- Tree representation of constituency structure. Phrase labels.
- CFG definition: terminals, nonterminals, start symbol, productions
- derivations and language of a CFG.
- derivation trees (vs. derived string).
- Regular grammars / languages(*)
- Complexity classes.
  - "Chomsky hierarchy"
  - Center embeddings as an example of a non-regular phenomenon.
  - Cross-serial dependencies as an example of a non-context-free phenomenon.
- Probabilitistic context free grammars (PCFG):
  - Probability of a tree vs. probability of a sentence.
  - Maximum likelihood estimates from treebanks.
  - Computing the most likely parse tree using CKY.
- Recognition (membership checking) problem vs. parsing
- Top-down vs. bottom-up parsing.
- **CKY parser:**
  - bottom-up approach.
  - Chomsky normal form.
  - Dynamic programming algorithm. (know the algorithm and required data structure: CKY parse table). Split position.
  - Backpointers.
  - Parsing with PCFGs

Dependency parsing (J&M 3rd ed. Ch 14.1-14.5
Supplementary material: Küber, McDonald, and Nivre (2009): Dependency Parsing, Ch.1 to 4.2)

- Transition based dependency parsing:
  - States (configuration): Stack, Buffer, Partial dependency tree
  - Transitions (Arc-standard system): Shift, Left-Arc, Right-Arc
  - Predicting the next transition using discriminative classifiers.
    - Feature definition (address + attribute)
  - Training the parser from a treebank:
    - Oracle transitions from annotated dependency tree.
  - Arc-eager system (*)
- Graph-based dependency parsing (* know the general idea: each word needs to get a head, start with completely connected graph, score each edge, keep the highest scoring edges)

Machine Learning (Some textbook references below)

- Generative vs. discriminative algorithms
- Supervised learning. Classification vs. regression problems, multi-class classification
- Loss functions: Least squares error. Classification error.
- Training vs. testing error. Overfitting.
- Linear Models.
  - activation function.
  - perceptron learning algorithm (*)
  - linear separability and the XOR problem. (*)
- Feature functions
- Log-linear / maximum entropy models (J&M 3rd. ed. ch. 7, J&M 2nd ed. ch. 6.6)
  - Log-likelihood of the model on the training data.
  - Simple gradient ascent.
  - Regularization (*)
  - MEMM (Maximum entropy markov models):
    - POS tagging with MEMMs.
- Feed-forward neural nets (J&M 3rd. ed. ch. 8, as well as Goldberg Ch 3. & 5.1)
  - Multilayer neural nets.
  - Different activation functions (sigmoid, ReLU, tanh)
  - Computation Graph.
  - Softmax.
  - Backpropagation (*) (know the idea of propagating back partial derivatives of the loss with respect to each weight -- you won't have to do calculus on the final).
  - Neural language model (Goldberg Ch. 9.4, J&M SLP 3rd. edition Ch. 6.2-6.5)

Vector-based Lexical Semantics (J&M 3rd ed. ch 15 & 16, not in 2nd ed.)

- Distributional hypothesis
- Co-occurrence matrix
- Distance/Similarity metrics (Euclidean distance, cosine similarity)
- Parameters of Distributional Semantic Models
  - Preprocessing, term definition, context definition, feature weighting (TF-IDF), normalization, dimensionality reduction (SVD (*)), similarity/distance measure
- Semantic similarity and relatedness (paradigmatic vs. syntagmatic relatedness)
  - Effect of context size on type of relatedness.
- Sparse vs. Dense Vectors
- One-hot representation (of words, word-senses, features)
- Word Embeddings
- Pre-training
- Word2Vec embeddings using a neural network. (Goldberg Ch. 10.1-10.4)
  - Skip-gram model
  - CBOW continuous bag of words
  - softmax objective, negative sampling

Symbolic Lexical Semantics (J&M 3rd ed. ch 17, J&M 2nd ed. ch 19.1-19.3 and 20.1-20.4)

- Word senses, lexemes, homonymy, polysemy, metonymy, zeugma.
- WordNet, synsets
- lexical relations (in WordNet)
  - synonym, antonym
  - hypernym, hyponym
  - meronym
- Word-sense disambiguation:
  - Supervised learning approach and useful features.
  - Lesk algorithm (J&M 3rd ed. ch. 17.6)
- Lexical substitution task

Recurrent Neural Nets

- Recurrent neural nets. (Goldberg Ch. 14.1-14.5)
  - Basic RNN concept, hidden layer as state representation.
  - Common usage patterns:
    - Acceptor / Encoder
    - Transducer
      - Transducer as generator
      - Conditioned transduction (encoder-decoder)
      - image captioning (*)
  - Backpropagation through time (BPTT) (*)
  - LSTMs (*)
  - Bidirectional RNNs
  - Stacked/Layered/Deep RNNs
  - Encoder/Decoder and Sequence to Sequence
    - Attention mechanism
      - Context vector
      - alignment score and attention weights
      - attention as a "lookup" operation (queries/keys/values)
- ELMo
  - contextualized word representations
  - model architecture, bi-Language Model training objective
  - transfer learning / pre-training
  - applying ELMo vectors to downstream tasks

Transformer Based Models (not in textbook)

- Transformer architecture (contrast to RNNs)
  - self attention (between encoder layers or decoder layers)
  - cross-attention (decoder pays attention to encoder representations)
  - scaled dot product attention function
  - multi-head attention
  - Encoder layers
  - Decoder layers
    - masked attention
  - Positional encoding (*)
- GPT
  - uses transformer decoder layers -> there is only self-attention
  - LM training objective
  - Application of GPT to classification problems
  - Representing various single/two sentence classification problems as input for GPT.
  - Fine tuning (weighted secondary objective)
  - Zero- and few-shot learning / prompting
- BERT
  - uses transformer encoder layers (-> no cross-attention)
  - input representation [CLS], [SEP], [MASK]
  - significance of the [CLS] token
  - masked language model training objective
  - next sentence prediction training objective
  - applying BERT to sequence labeling and prediction tasks (and fine tuning)

Statistical MT (Some sections of J&M 3rd ed. Ch. 10, also see Michael Collins' notes on IBM M2 here: http://www.cs.columbia.edu/~mcollins/ibm12.pdf)

- Challenges for MT, word order, lexical divergence
- Vauquois triangle.
- Faithfulness vs. Fluency
- Parallel Corpora
- Noisy Channel Model for MT
- Word alignments
- IBM Model 2:
  - Alignment variables and model definition (t = translation parameters and q = alignment parameters).
  - EM training for Model 2 (*)
- Phrase-based MT (*)
- MT Evaluation:
  - BLEU Score (precision of different ngrams)
  - Brevity penalty (*)

Semantic Roles / Semantic Role Labeling (J&M Ch. 22)

- General concept of thematic/semantic roles vs. grammatical functions.
- Alternation patterns and verb classes (you don't have to remember specific alternation patterns)
- VerbNet
- PropBank:
  - ARGx argument roles and ARGMs
  - ARG0 = proto agent, ARG1 = proto patient (*)
  - framesets
- Frame Semantics
  - Frame, Frame Elements (specific to each frame)
  - Valence Patterns
  - FrameNet:
    - Frame definitions
    - Lexical Units
    - Example annotations to illustrate valence patterns.
    - Frame-to-frame relations and frame-element relations (*) (you do not need to remember an exhaustive list of these relations).
  - Semantic Role Labeling:
    - Steps of the general syntax-based approach
      - Target identification, semantic role/frame element identification and labeling.
    - Features for semantic role role/FE identification and labeling. (*) (you do not need to remember an exhaustive list, but have a general sense of which features are important for this task).
      - Parse-tree path features
    - RNN-based SRL (BIO sequence tagging)
    - Span-graph based approach (*)